

Hydrochemical evolution and spatio-temporal dynamics of irrigation water quality in the Bay of Algiers aquifer (Algeria)

Evoluzione idrochimica e dinamica spatio-temporale della qualità delle acque irrigue nell'acquifero della Baia di Algeri (Algeria)

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Abstract

This study assesses the spatio-temporal evolution of the quality of groundwater intended for irrigation in the coastal aquifer of the Bay of Algiers (northern Algeria) during three sampling campaigns conducted in 2006, 2020 and 2025. An integrated approach combining physicochemical parameters, the main irrigation water quality indices, hydrochemical diagrams, multivariate statistical analysis and geographic information systems was used to characterise the controlling hydrochemical processes and assess the suitability of the water for irrigation. The results reveal a progressive deterioration in water quality, marked by increases in electrical conductivity, total dissolved solids, sodium and chloride. The irrigation water quality index decreased from 64.82 in 2006 to 55.19 in 2020 and then to 36.08 in 2025, indicating a progressive reduction in irrigation suitability. The Piper and Gibbs diagrams, together with hydrochemical indices and spatial mapping, highlight an intensification of salinisation processes driven by the combined effects of water–rock interaction, evaporation–crystallisation, an increasing influence of seawater intrusion and anthropogenic pressures. These findings underline the need for regular monitoring and sustainable management strategies to preserve groundwater resources and limit salinisation risks in coastal aquifers.

Riassunto

Questo studio valuta l'evoluzione spaziale e temporale della qualità delle acque sotterranee per uso irriguo nell'acquifero costiero della Baia di Algeri (Algeria settentrionale) nel corso di tre campagne di campionamento condotte nel 2006, 2020 e 2025. È stato adottato un approccio integrato che combina parametri fisico-chimici, i principali indici di qualità dell'acqua irrigua, diagrammi idrochimici, analisi statistica multivariata e sistemi informativi geografici, al fine di caratterizzare i processi idrochimici e valutare l'idoneità dell'acqua all'irrigazione. I risultati rivelano un progressivo deterioramento della qualità dell'acqua, caratterizzato da un aumento della conducibilità elettrica, dei solidi disciolti totali, del sodio e dei cloruri. L'indice di qualità dell'acqua irrigua è diminuito da 64.82 nel 2006 a 55.19 nel 2020 e quindi a 36.08 nel 2025, indicando una progressiva riduzione dell'idoneità all'irrigazione. I diagrammi di Piper e di Gibbs, insieme agli indici idrochimici e alla cartografia spaziale, evidenziano un'intensificazione dei processi di salinizzazione, determinata dagli effetti combinati dei processi di interazione acqua-roccia ed evaporazione-cristallizzazione, di una crescente incidenza dell'intrusione marina e delle pressioni antropiche. Questi risultati evidenziano la necessità di un monitoraggio regolare e di strategie di gestione sostenibile al fine di preservare le risorse idriche sotterranee e di limitare i rischi di salinizzazione negli acquiferi costieri.

Introduction

Agriculture accounts for more than 70% of global freshwater withdrawals (FAO, 2025). Rising food demand is increasing the need for irrigation water, making groundwater a strategic resource (Gugulothu et al., 2022; Subba Rao et al., 2017; Zaman et al., 2018). However, intensive groundwater exploitation, combined with anthropogenic pressures, promotes aquifer overexploitation and the progressive deterioration of water quality (Adimalla & Taloor, 2020).

The quality of groundwater intended for irrigation is mainly controlled by the hydrogeochemical processes governing water–rock interactions, under the influence of lithology, recharge, residence time and ion exchange (Gugulothu et al., 2022; Subba Rao et al., 2017). Anthropogenic activities—in particular urbanisation, industrialisation, inadequate sanitation and the intensive use of fertilisers—also contribute to its chemical degradation (Li, 2016; Masoud & Abu El-Magd, 2022; Noori et al., 2021; Wagh et al., 2019).

The assessment of groundwater quality for irrigation often relies on an integrated approach combining physicochemical parameters, water quality indices, multivariate statistical methods and geographic information systems (GIS). These tools make it possible to characterise hydrogeochemical processes, assess the suitability of water for irrigation and analyse its spatio-temporal variability (Rawat et al., 2018; Gaagai et al., 2023). Among the most widely used approaches, the irrigation water quality index (IWQI) provides a synthetic assessment of the risks associated with salinity, sodicity and ionic toxicity (Meireles et al., 2010).

Coastal aquifers are among the groundwater resources most vulnerable to quality deterioration. Aquifer overexploitation, reduced natural recharge and sea-level rise promote the advance of seawater intrusion, leading to a progressive enrichment of dissolved salts in groundwater. This salinisation is one of the main factors limiting the use of groundwater for irrigation and may permanently affect the agricultural productivity of coastal areas (Solangi et al., 2019). In this context, GIS provides a complementary tool for visualising the spatial distribution of quality parameters and indices and for identifying the most vulnerable areas.

In Algeria, studies of groundwater quality have focused mainly on resources intended for drinking-water supply, whereas studies addressing irrigation remain limited. Available research reports a progressive deterioration in water quality in several aquifers, mainly associated with salinisation and seawater intrusion (Selmane et al., 2023; Gaagai et al., 2023; Yahiaoui et al., 2023), highlighting the need for a comprehensive assessment of resources intended for irrigation.

The coastal aquifer of the Bay of Algiers, located in the Mitidja Plain in northern Algeria, is a strategic resource for drinking-water supply and irrigation. Numerous studies have characterised its hydrogeological functioning, chemical quality and the advance of seawater intrusion (Aziez et al., 2025; Djoudar Hallal & Toubal, 2008; Takorabt, 2011; Yahiaoui et al., 2023). However, studies specifically devoted to assessing its suitability for irrigation and its spatio-

temporal evolution remain very limited, even though the progressive salinisation of the aquifer poses a long-term threat to agricultural uses.

This study has four objectives. The first two are to: (a) characterise groundwater hydrochemical evolution and the geochemical controls on its composition; and (b) assess its irrigation suitability using physicochemical parameters and quality indices. The other objectives are to: (c) analyse spatio-temporal changes in 2006, 2020 and 2025; and (d) map the spatial distribution of the parameters and indices using GIS. To our knowledge, this study provides the first multi-decadal (2006–2025) assessment of the irrigation suitability of the coastal aquifer of the Bay of Algiers, integrating the IWQI, irrigation indices, hydrochemical diagrams, multivariate statistics and GIS. It also complements the general trend analysis proposed by Ferdjouni et al. (2025) through an approach centred on hydrochemical processes and agronomic issues.

Materials and methods

Study area

The study area is located in the Algiers Sahel, north of the Mitidja Plain, and extends along the Bay of Algiers, from the mouth of the El Harrach River to Cape Bordj El Bahri. This coastal strip forms the natural outlet of the eastern Mitidja hydrological system, where the main surface-water and groundwater flows converge before discharging into the Mediterranean Sea. Owing to its strategic position at the interface between continental aquifers and the marine environment, this area is particularly sensitive to saline intrusion and anthropogenic pressures (Fig. 1).

The Mitidja Plain, in northern Algeria, covers approximately 1,450 km² (between 36°22'30" and 36°48'58" N, and 2°32'07" and 3°28'14" E). It has a generally flat topography, bordered to the south by the foothills of the Blidean Atlas and to the north by the Algiers Sahel, where the present study area is located. The climate is Mediterranean with a semi-arid tendency, characterised by mild, wet winters and hot, dry summers. The mean annual temperature is approximately 18.5 °C, annual precipitation ranges from 500 to 780 mm, and potential evapotranspiration is high (970–1,250 mm/year).

Hydrologically, the eastern Mitidja Plain is drained mainly by the El Harrach, El Hamiz and Réghaïa rivers, which rise in the highlands of the Blidean Atlas before crossing the plain and flowing into the Bay of Algiers. These rivers play a major role in aquifer recharge and in the transfer of water and contaminants towards the coastal zone. Geologically, the Mitidja Plain is a subsiding basin filled with Plio-Quaternary deposits derived from erosion of the Blidean Atlas. This evolution has produced two superposed aquifer systems. The first is the Astian aquifer (Upper Pliocene), a deep, confined aquifer located at depths ranging from approximately 200 m to more than 400 m. It consists of marly-sandy, limestone and sandstone formations overlying an impermeable clayey basement. The second is the Quaternary alluvial aquifer of the Mitidja Plain, which is unconfined and shallow, with

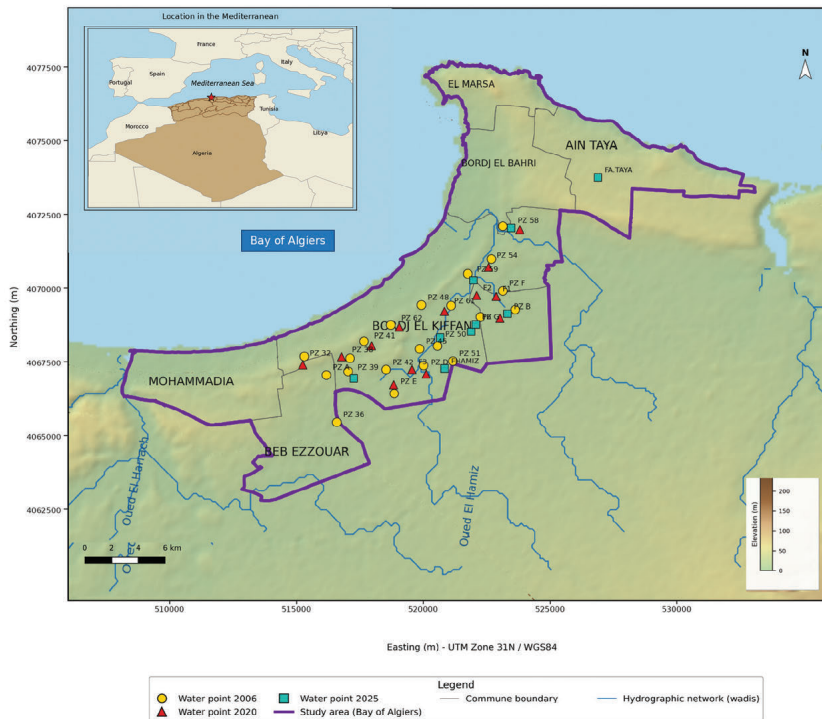


Fig. 1 - Location of the Bay of Algiers aquifer and groundwater sampling points used in 2006, 2020 and 2025. Coordinate reference system: WGS 84 / UTM zone 31N.

Fig. 1 - Ubicazione dell'acquifero della Baia di Algeri e punti di monitoraggio delle acque sotterranee negli anni 2006, 2020 e 2025. Sistema di coordinate: WGS84 / UTM zona 31N.

an average thickness of 100–130 m. It consists of pebbles, gravels and sands interbedded with clay layers. In the western Mazafran sector, it becomes locally confined beneath a silty cover. The two reservoirs are separated by the impermeable clays of the El Harrach Formation (Villafranchian).

The Mitidja Plain is one of Algeria's most important agricultural regions, with market gardening (tomatoes, lettuce, cabbage, and eggplant), arboriculture (citrus fruits, pomegranate, apricot and olive), cereal growing (durum wheat and maize) and viticulture (grapevines). The aquifers also provide a substantial proportion of the region's drinking-water supply, irrigation water and industrial water. This high demand is placing increasing pressure on groundwater resources, resulting in aquifer overexploitation, hydrodynamic disturbances, seawater intrusion—particularly affecting the Bordj El Kiffan and Stamboul sectors—and deterioration in water quality caused by fertiliser inputs and inadequately treated urban and industrial discharges. As an interface between the Mitidja aquifers and the Mediterranean Sea, the Algiers Sahel therefore provides a natural laboratory for studying groundwater-seawater interactions and assessing saline-intrusion processes and anthropogenic impacts on coastal water resources.

Data and analytical methods

The data used in this study were obtained from three sampling campaigns conducted in 2006, 2020 and 2025. Samples were collected during the low-water period (dry season) from 21, 13 and 9 water points, respectively (Fig. 1), giving a total of 43 samples. All monitoring wells exploit the upper alluvial aquifer of the Mitidja Plain, which is a shallow, unconfined aquifer. The physicochemical data were obtained

from the national groundwater monitoring network managed by the National Agency for Hydraulic Resources (ANRH) in Blida. Twelve physicochemical parameters were selected to characterise the water and assess its suitability for irrigation: pH, electrical conductivity (EC), calcium (Ca^{2+}), magnesium (Mg^{2+}), potassium (K^{+}), sodium (Na^{+}), chloride (Cl^{-}), sulphate (SO_4^{2-}), bicarbonate (HCO_3^{-}), nitrate (NO_3^{-}), total dissolved solids (TDS) and total hardness (TH).

The number of accessible water points decreased during the study period, from 21 in 2006 to 13 in 2020 and then to 9 in 2025. Several monitoring wells became unusable because of urbanisation, abandonment or excessive salinisation. To ensure comparability between campaigns, the three datasets were interpreted for the same coastal sectors of the aquifer. The temporal analysis was based on the statistical distribution (mean, dispersion and quality classes) of each variable rather than on point-by-point comparisons. The progressive reduction in the monitoring network is nevertheless recognised as a limitation, and the 2025 results, based on a smaller sample, are interpreted with due caution.

The ionic balance error (%CBE, Eq. 1) was calculated and confirmed that all samples had a value within the $\pm 5\%$ range :

$$\%CBE = \frac{\sum cations - \sum anions}{\sum cations + \sum anions} \cdot 100 \quad (1)$$

The mapping of groundwater quality and the identification of vulnerable zones were carried out using ArcGIS 10.5 software by applying the Inverse Distance Weighted (IDW) interpolation method. Descriptive statistical analyses, as well as the generation of Piper, Wilcox, and USSSL hydrochemical diagrams, were performed using Diagrams software, while Pearson correlation matrices were processed using XLSTAT.

Hydrochemical characteristics and irrigation indices

Groundwater hydrochemistry was assessed using the main physicochemical parameters (pH, EC, Ca^{2+} , Mg^{2+} , K^+ , Na^+ , Cl^- , SO_4^{2-} , HCO_3^- , NO_3^- , TDS and TH) determined during the three sampling campaigns. Total dissolved solids (TDS) were estimated from electrical conductivity using the relationship $\text{TDS} \approx 0.64 \times \text{EC}$ (where EC is expressed in $\mu\text{S}/\text{cm}$). Irrigation suitability was evaluated using SAR, Na%, RSC, RSBC, MH, KR, PI, PS and IWQI, while hydrochemical evolution was assessed using CAI-1, CAI-2, r_1 and r_2 . The corresponding equations, classification criteria and references are provided in Supplementary Table S1. The chloro-alkaline indices (CAI-1 and CAI-2) and Soltan ratios (r_1 and r_2) were calculated to assess ion exchange, salinisation processes and the origin of mineralisation, according to the interpretation criteria of Schoeller (1965) and Soltan (1999). Positive CAI values indicate the exchange of dissolved Na^+ and K^+ for Ca^{2+} and Mg^{2+} fixed in the aquifer matrix, whereas negative values indicate the reverse exchange. For r_1 , values greater than 1 characterise a Na– HCO_3 facies, whereas values lower than 1 indicate a Na– SO_4 facies. For r_2 , values lower than 1 indicate deep meteoric percolation, while values greater than 1 indicate shallow meteoric percolation.

The Piper diagram (Piper, 1944) was used to identify the main hydrochemical facies of groundwater, while the Gibbs diagram (Gibbs, 1970) was used to identify the dominant mechanisms controlling water chemistry (precipitation, water–rock interaction and evaporation). USSL (EC–SAR) and Wilcox (%Na–EC) diagrams were used to assess the suitability of water for irrigation according to salinity and sodicity risks.

Irrigation water quality index (IWQI)

The IWQI is a widely used multicriteria tool for assessing the overall suitability of water resources for agriculture (Gaagai et al., 2023; Meireles et al., 2010). Unlike conventional hydrochemical approaches, which assess risks separately, the IWQI combines the effects of several key chemical parameters into a single numerical value. It simultaneously incorporates the main variables responsible for salinity, sodicity and ionic toxicity risks, namely EC, SAR, Na^+ , Cl^- and HCO_3^- (Abbasnia et al., 2018). Concentrations expressed in mg/L are first converted to meq/L, after which the index is calculated in two successive steps (Meireles et al., 2010).

Step 1 - Individual quality score (q_i). Each parameter is assigned a quality score (q_i) ranging from 0 to 100 according to its measured concentration and its level of restriction for irrigation:

$$q_i = q_{i\max} - \frac{(x_{ij} - x_{\inf}) \cdot q_{i\text{amp}}}{x_{\text{amp}}} \quad (2)$$

where $q_{i\max}$ is the maximum value of the quality class considered, x_{ij} is the measured value of the parameter, $x_{\inf}$ is the lower limit of the class, $q_{i\text{amp}}$ is the class amplitude and x_{amp} is the amplitude of the parameter class. For the last class

of each variable, the maximum measured value was taken as the upper limit. The criteria used to calculate the q_i scores are given in Table 1.

Tab. 1 - Limiting values used to compute the individual quality score (q_i).

Tab. 1 - Valori soglia usati per il calcolo del punteggio di qualità individuale (q_i).

q_i (%)	EC ($\mu\text{S}/\text{cm}$)	SAR	Na^+ (meq/L)	Cl^- (meq/L)	HCO_3^- (meq/L)
85–100	20–750	<3	2–3	<4	1–1.5
60–85	750–1500	3–6	3–6	4–7	1.5–4.5
35–60	1500–3000	6–12	6–9	7–10	4.5–8.5
0–35	<200 or >3000	>12	<2 or >9	>10	<1 or >8.5

Step 2 - Overall index. The weighted sum of the individual scores provides the IWQI, with the weights reflecting the relative importance of each parameter in relation to salinity, sodicity or ionic toxicity risks (with $\sum w_i = 1$):

$$\text{IWQI} = \sum_{i=1}^n q_i \cdot w_i \quad (3)$$

IWQI values range from 0 to 100, with higher values indicating better quality and greater suitability for irrigation. According to the classification of Meireles et al. (2010), the water was assigned to the classes presented in Table 2.

Tab. 2 - IWQI classes and irrigation restrictions (after Meireles et al., 2010).

Tab. 2 - Classi dell'indice di qualità delle acque irrigue (IWQI) e restrizioni per l'irrigazione (da Meireles et al., 2010).

IWQI value	Class	Water-use restriction
85–100	C5	No restriction
70–85	C4	Low restriction
55–70	C3	Moderate restriction
40–55	C2	High restriction
<40	C1	Severe restriction

Statistical analysis and spatial interpolation

Pearson correlation coefficients were calculated separately for the 2006, 2020 and 2025 campaigns. Two-sided significance was assessed at $\alpha = 0.05$ and $\alpha = 0.01$. Only statistically significant associations are interpreted as evidence of a relationship. Total dissolved solids (TDS), estimated from electrical conductivity, were excluded from the matrices to avoid a mathematically induced correlation. Because sample size decreased from 21 observations in 2006 to 13 in 2020 and 9 in 2025, comparisons of correlation magnitude and significance among campaigns were interpreted cautiously; reduced statistical power in the later campaigns can cause genuine associations to remain non-significant. Spatial maps were generated in ArcGIS 10.5 using inverse distance weighting (IDW).

Results

This section presents the physicochemical, hydrochemical and statistical results used to characterise changes in groundwater quality and irrigation suitability between 2006 and 2025. Descriptive statistics for the three campaigns are

presented in Table 3; the complementary plot of temporal mean values is provided as Supplementary Figure S1.

Physico-chemical parameters

Unlike pH, which shows stable values within the recommended limits for irrigation (Supplementary Table S1 and 3) indicating that this parameter is not a limiting factor-electrical conductivity (EC) is a major indicator of salinity risk in irrigation water (Ayers & Westcot, 1985). The results show

a progressive increase in mineralisation. Mean EC rose from $1,420 \pm 580 \mu\text{S}/\text{cm}$ in 2006 to $2,410 \pm 1,570 \mu\text{S}/\text{cm}$ in 2020 and $3,980 \pm 1,710 \mu\text{S}/\text{cm}$ in 2025 (Table 3), indicating an intensification of salinity stress. Spatial distribution confirms this trend: in 2006, permissible quality dominated the study area, while areas of doubtful quality remained localised. In 2020, doubtful quality spread mainly towards the central and eastern sectors. By 2025, it had become predominant across almost the entire study area, except for an area of good quality around Ain Taya (Fig. 2).

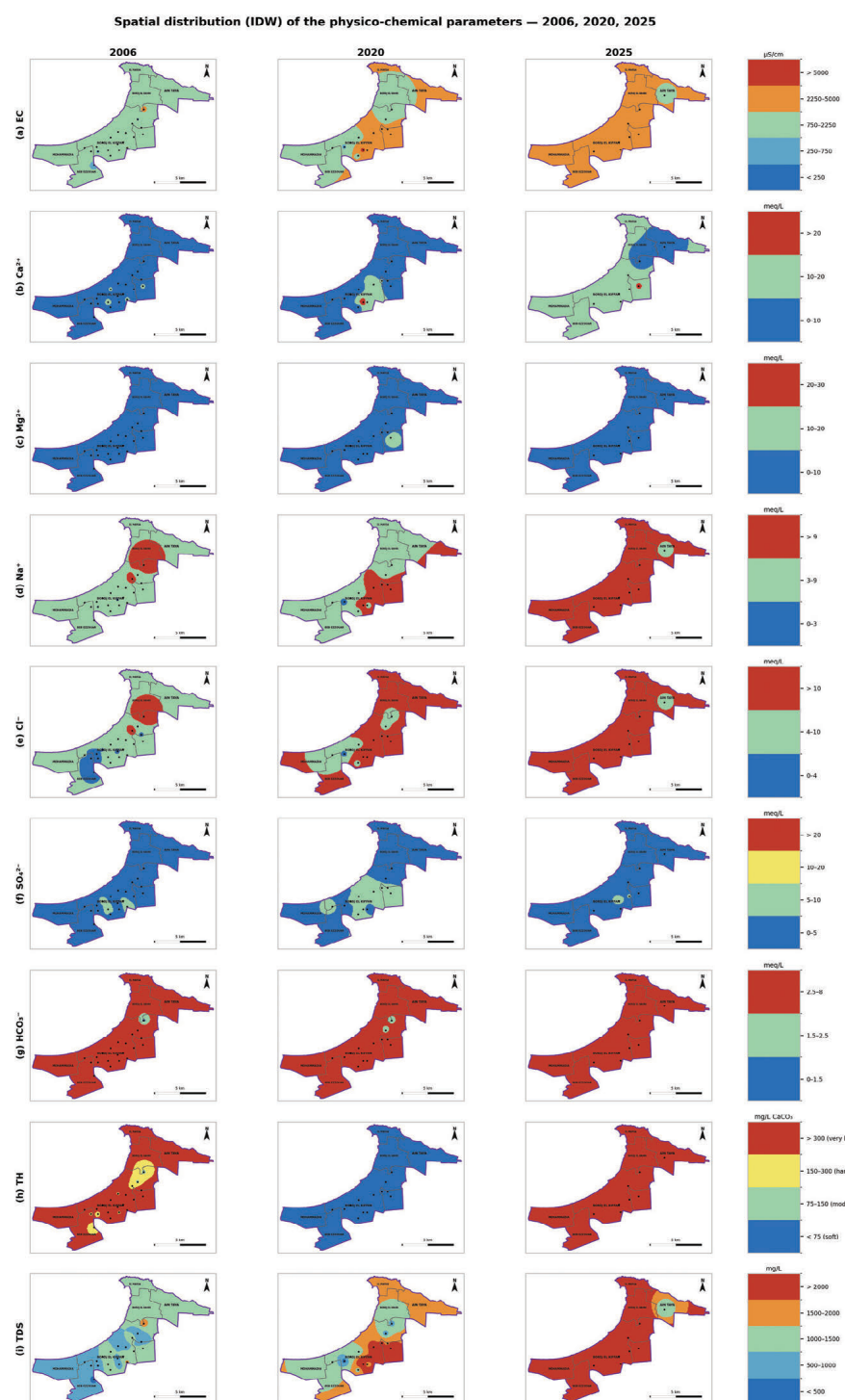


Fig. 2 - Spatial distribution (inverse distance weighting interpolation) of the main physicochemical parameters of groundwater in 2006, 2020 and 2025: (a) EC, (b) Ca^{2+} , (c) Mg^{2+} , (d) Na^{+} , (e) Cl^{-} , (f) SO_4^{2-} , (g) HCO_3^{-} , (h) TH and (i) TDS.

Fig. 2 - Distribuzione spaziale (interpolazione inversa della distanza) dei principali parametri fisico-chimici nelle acque sotterranee negli anni 2006, 2020 e 2025: (a) EC, (b) Ca^{2+} , (c) Mg^{2+} , (d) Na^{+} , (e) Cl^{-} , (f) SO_4^{2-} , (g) HCO_3^{-} , (h) TH and (i) TDS.

Tab. 3 - Descriptive statistics of the physicochemical parameters of groundwater in 2006, 2020 and 2025 (Min; Max; Mean $\pm$ SD).Tab. 3 - Statistiche descrittive dei parametri fisico-chimici nelle acque sotterranee negli anni 2006, 2020 e 2025 (minimo, massimo, media $\pm$ deviazione standard).

Parameter	Unit	2006			2020			2025		
		Min	Max	Mean $\pm$ SD	Min	Max	Mean $\pm$ SD	Min	Max	Mean $\pm$ SD
pH	—	500	8.00	7.50 $\pm$ 0.26	7.30	8.30	7.97 $\pm$ 0.31	7.50	8.00	7.79 $\pm$ 0.16
EC	μ S/cm	5000	2430	1420 $\pm$ 0580	500	6450	2410 $\pm$ 1570	1490	7200	3980 $\pm$ 1710
Ca ²⁺	meq/L	0.68	13.18	6.26 $\pm$ 3.81	1.20	31.14	8.44 $\pm$ 8.04	2.68	21.96	13.76 $\pm$ 7.32
Mg ²⁺	meq/L	0.76	5.98	2.79 $\pm$ 1.52	1.38	19.75	5.93 $\pm$ 4.86	1.98	7.90	4.94 $\pm$ 1.72
Na ⁺	meq/L	2.26	20.92	5.93 $\pm$ 4.32	1.39	22.84	8.54 $\pm$ 5.71	4.83	43.50	19.73 $\pm$ 12.47
K ⁺	meq/L	0.00	0.20	0.04 $\pm$ 0.06	0.00	0.51	0.13 $\pm$ 0.15	0.05	0.20	0.11 $\pm$ 0.05
Cl ⁻	meq/L	1.80	21.86	6.26 $\pm$ 5.03	2.17	49.49	13.64 $\pm$ 13.35	4.68	57.39	28.51 $\pm$ 16.82
SO ₄ ²⁻	meq/L	0.06	9.62	3.32 $\pm$ 2.95	0.62	9.83	4.89 $\pm$ 2.53	0.31	6.99	2.72 $\pm$ 2.51
HCO ₃ ⁻	meq/L	1.35	7.16	4.42 $\pm$ 1.42	1.75	5.76	3.95 $\pm$ 1.28	2.60	7.00	4.66 $\pm$ 1.32
NO ₃ ⁻	meq/L	0.00	1.47	0.60 $\pm$ 0.58	0.00	1.93	0.68 $\pm$ 0.60	0.00	3.06	1.08 $\pm$ 0.92
TH	°f	11.52	86.79	45.51 $\pm$ 24.37	19.0	200	72.38 $\pm$ 49.63	44.33	140.00	93.93 $\pm$ 37.63
TDS	mg/L	350	1701	994.66 $\pm$ 403.74	350	4515	1684.31 $\pm$ 1100.20	1042.30	5040.00	2787.40 $\pm$ 1198.74

Total dissolved solids (TDS) and total hardness (TH) confirm the increase in groundwater mineralisation between 2006 and 2025 (Table 3; Fig. 2). TDS increased from 994.66 $\pm$ 403.74 mg/L in 2006 to 1,684.31 $\pm$ 1,100.20 mg/L in 2020 and 2,787.40 $\pm$ 1,198.74 mg/L in 2025. High concentrations progressively expanded towards the central and eastern sectors. TH also increased, from 45.51 $\pm$ 24.37 °f to 72.38 $\pm$ 49.63 °f and then to 93.93 $\pm$ 37.63 °f, with an expansion of high values in 2025. These changes are consistent with the increase in EC and confirm the strengthening of mineralisation during the study period.

Major ions

The spatial distributions of the ions show differentiated hydrochemical evolution between 2006 and 2025 (Fig. 2). Na⁺ and Cl⁻ show the most marked changes. In 2006, their high concentrations were mainly located at Bordj El Bahri and El Marsa. In 2020, high Na⁺ concentrations occurred particularly at Bordj El Kiffan, while Cl⁻ concentrations spread widely, reaching 49.49 meq/L. In 2025, Na⁺ concentrations >9 meq/L covered El Marsa, Bordj El Bahri, Bordj El Kiffan, Mohammadia and Bab Ezzouar, while Cl⁻ concentrations remained high, with a maximum value of 57.39 meq/L. Ca²⁺ concentrations showed fluctuating behaviour, with maxima of 13.18, 31.14 and 21.96 meq/L in 2006, 2020 and 2025, respectively; the highest concentrations were mainly located at Bordj El Kiffan. Mg²⁺ concentrations remained mostly <10 meq/L, except for a localised anomaly at Bordj El Kiffan in 2020. HCO₃⁻ shifted from a predominance of concentrations above 2.5 meq/L in 2006–2020 to a predominance of the 1.51–2.5 meq/L class in 2025, with maxima of 7.16, 5.76 and 7 meq/L, respectively. SO₄²⁻ remained dominated by low concentrations ($\leq$ 5.2 meq/L), despite a localised expansion of intermediate concentrations at Bordj El Kiffan in 2020. Thus, Na⁺ and Cl⁻ indicate the most substantial spatial changes, whereas Ca²⁺ fluctuated and Mg²⁺, HCO₃⁻ and SO₄²⁻ remained generally more stable.

Correlation analysis

The correlation matrices (Supplementary Tables S2, S3 and S4) were interpreted chronologically and only statistically significant coefficients are discussed. In 2006 (n = 21), EC was positively associated with Ca²⁺ (r = 0.640, p < 0.01), Mg²⁺ (r = 0.451, p < 0.05) and Na⁺ (r = 0.565, p < 0.01). Significant positive associations were also found with Cl⁻ (r = 0.682, p < 0.01), SO₄²⁻ (r = 0.628, p < 0.01), NO₃⁻ (r = 0.470, p < 0.05) and TH (r = 0.644, p < 0.01). The strongest relationships included Ca²⁺-TH (r = 0.966), SO₄²⁻-TH (r = 0.909), Ca²⁺-SO₄²⁻ (r = 0.855) and Na⁺-Cl⁻ (r = 0.946; all p < 0.01). In 2020 (n = 13), EC was strongly related to Ca²⁺, Na⁺, Cl⁻, SO₄²⁻ and TH (r = 0.819–0.978; all p < 0.01), and Na⁺-Cl⁻ remained strongly correlated (r = 0.959, p < 0.01). In 2025 (n = 9), EC was significantly related to Ca²⁺ (r = 0.675, p < 0.05), Na⁺ (r = 0.921, p < 0.01), Cl⁻ (r = 0.973, p < 0.01) and TH (r = 0.736, p < 0.05). Other significant relationships were Na⁺-Cl⁻ (r = 0.951, p < 0.01), Ca²⁺-TH (r = 0.973, p < 0.01) and pH-Mg²⁺ (r = 0.756, p < 0.05). The apparently weaker and fewer significant relationships in 2025 should not be interpreted by themselves as proof of changed hydrochemical processes. They are also consistent with the lower power and greater coefficient instability expected from the progressive reduction in sample size. Non-significant coefficients, including the negative NO₃⁻ associations in 2025, are therefore not interpreted.

Irrigation water quality indices

Table 4 summarizes the descriptive statistics of the main irrigation indices for the three campaigns.

SAR values ranged from 1.02 to 18.32 in 2006 (3.46 $\pm$ 3.91), from 0.99 to 5.12 in 2020 (3.19 $\pm$ 1.37), and from 2.30 to 15.82 in 2025 (6.55 $\pm$ 4.38) (Table 4). Following a slight decrease between 2006 and 2020, SAR increased in 2025. Values below 10 remained predominant, representing 90.5%,

Tab. 4 - Descriptive statistics of the irrigation suitability indices (Min; Max; Mean $\pm$ SD).Tab. 4 - Statistiche descrittive degli indici di idoneità all'irrigazione (minimo, massimo, media $\pm$ deviazione standard).

Index	2006			2020			2025		
	Min	Max	Mean $\pm$ SD	Min	Max	Mean $\pm$ SD	Min	Max	Mean $\pm$ SD
SAR	1.02	18.32	3.46 $\pm$ 3.91	0.99	5.12	3.19 $\pm$ 1.37	2.30	15.82	6.55 $\pm$ 4.38
%Na	19.60	89.01	40.26 $\pm$ 19.06	23.97	62.91	38.50 $\pm$ 11.78	35.71	78.09	48.41 $\pm$ 13.27
RSC	-12.49	0.85	-4.63 $\pm$ 4.28	-30.03	-1.48	-9.48 $\pm$ 7.99	-22.05	-4.46	-14.04 $\pm$ 7.14
PI	0.32	0.94	0.56 $\pm$ 0.20	0.36	0.78	0.49 $\pm$ 0.12	0.44	0.82	0.55 $\pm$ 0.12
KR	0.23	8.02	1.12 $\pm$ 1.75	0.31	1.66	0.68 $\pm$ 0.40	0.55	3.55	1.14 $\pm$ 0.95
MH	15.34	77.47	33.36 $\pm$ 14.91	9.43	71.21	46.35 $\pm$ 22.35	16.78	69.55	31.21 $\pm$ 19.67
PS	2.29	21.89	7.92 $\pm$ 5.10	3.59	28.50	10.77 $\pm$ 6.91	5.41	60.88	29.87 $\pm$ 17.12
RSBC	3.96	23.94	13.48 $\pm$ 5.73	4.60	49.69	19.26 $\pm$ 12.03	12.79	34.08	23.35 $\pm$ 8.07
IWQI	34.64	77.82	64.82 $\pm$ 11.23	25.95	84.13	55.19 $\pm$ 16.53	15.17	67.17	36.08 $\pm$ 18.09

100% and 77.8% of samples, respectively. Higher values were localised mainly in the Bordj El Bahri sector (Fig. 3), indicating a spatially limited increase in sodicity risk in 2025.

The sodium percentage (%Na) shifted from $40.26 \pm 19.06\%$ in 2006 to $38.50 \pm 11.78\%$ in 2020, and then to $48.41 \pm 13.27\%$ in 2025 (Table 4; Fig. 3), showing an increase in 2025. Spatially, moderate values dominated in 2006, particularly in Mohammedia, Bab Ezzouar, and Bordj El Kiffan, while higher values were observed in El Marsa and Aïn Taya, with a localised anomaly in Bordj El Bahri. In 2020, an increase in values extended towards the northern and northeastern sectors, with a localised anomaly in the Bordj El Kiffan–Bordj El Bahri sector. By 2025, higher values became dominant across the entire study area, while an anomaly persisted in Bordj El Bahri. The proportion of samples presenting values compatible with good to permissible irrigation suitability represented the majority (Supplementary Table S1), but the increase in %Na in 2025 indicates a degradation of sodic quality, consistent with the evolution of SAR, electrical conductivity, and chlorides.

RSBC increased from 13.48 ± 5.73 meq/L in 2006 to 19.26 ± 12.03 meq/L in 2020, and further to 23.35 ± 8.07 meq/L in 2025, whereas the minimum value rose from 3.96 to 12.79 meq/L (Table 4). Spatial maps (Fig. 3) show a predominance of high values in the sectors of El Marsa, Aïn Taya, Bordj El Kiffan, Mohammedia, and Bab Ezzouar during 2006–2020, with an extension towards Bordj El Bahri in 2020. In 2025, intermediate values covered most of the study area, while higher values were localised primarily in the Bordj El Kiffan sector. Despite this spatial distribution, the proportion of samples with values exceeding 10 meq/L increased from 66.7% in 2006 to 76.9% in 2020, reaching 100% in 2025 (Supplementary Table S1), reflecting a generalised deterioration of water suitability for irrigation regarding the sodification hazard.

KR values ranged from 0.23 to 8.02 in 2006 (1.12 ± 1.75), from 0.31 to 1.66 in 2020 (0.68 ± 0.40), and from 0.55 to 3.55 in 2025 (1.14 ± 0.95) (Table 4; Fig. 3). Following a decrease in 2020, KR increased again in 2025, returning to a mean level close to that of 2006. The spatial distribution

showed high values principally in El Marsa, Aïn Taya, and Bordj El Bahri in 2006, predominantly in Bordj El Bahri in 2020, and subsequently in El Marsa, Bordj El Bahri, and Mohammedia in 2025. The proportion of samples with values below 1 remained the majority, accounting for 76.2%, 84.6%, and 77.8% in 2006, 2020, and 2025, respectively, whereas values above 1 concerned 23.8%, 15.4%, and 22.2% of the samples. This evolution highlights a localised degradation of water suitability regarding sodium in 2025, consistent with the trends observed for SAR and %Na.

Magnesium hazard (MH) averaged $33.36 \pm 14.91\%$ in 2006, $46.35 \pm 22.35\%$ in 2020, and $31.21 \pm 19.67\%$ in 2025 (Table 4; Fig. 3), showing an increase in 2020 followed by a decrease in 2025. The proportion of samples with values $>50\%$ reached 9.5%, 46.2%, and 22.2%, respectively. Spatially, these values were mainly localised in Bordj El Kiffan in 2006, then extended to Bordj El Bahri and Mohammedia in 2020. In 2025, they became more punctual, mainly in Bordj El Bahri and Bordj El Kiffan, while the majority of samples remained favourable for irrigation (Supplementary Table S1).

Potential salinity (PS) increased from 7.92 ± 5.10 meq/L in 2006 to 10.77 ± 6.91 meq/L in 2020, and to 29.87 ± 17.12 meq/L in 2025 (Table 4; Fig. 3). Mean values exceeded 5 meq/L during all three campaigns, and in 2025, the minimum value (5.41 meq/L) also exceeded this threshold. Spatially, high values—initially concentrated in El Marsa, Bordj El Bahri, Aïn Taya, Bordj El Kiffan, and Mohammedia—gradually extended across most of the study area during 2020–2025, notably towards Bab Ezzouar, reflecting a progressive degradation of water quality regarding potential salinity.

The permeability index (PI) averaged 0.56 ± 0.20 in 2006, 0.49 ± 0.12 in 2020, and 0.55 ± 0.12 in 2025 (Table 4). All samples belonged to the same category according to Doneen (1964), indicating unfavourable irrigation suitability. The spatial distribution remained homogeneous across the three campaigns (Fig. 3).

IWQI values ranged from 34.64 to 77.82 in 2006 (64.82 ± 11.23), from 25.95 to 84.13 in 2020 (55.19 ± 16.53) and from 15.17 to 67.17 in 2025 (36.08 ± 18.09) (Table 4; Fig. 3). These results reveal a marked progressive decline in irrigation

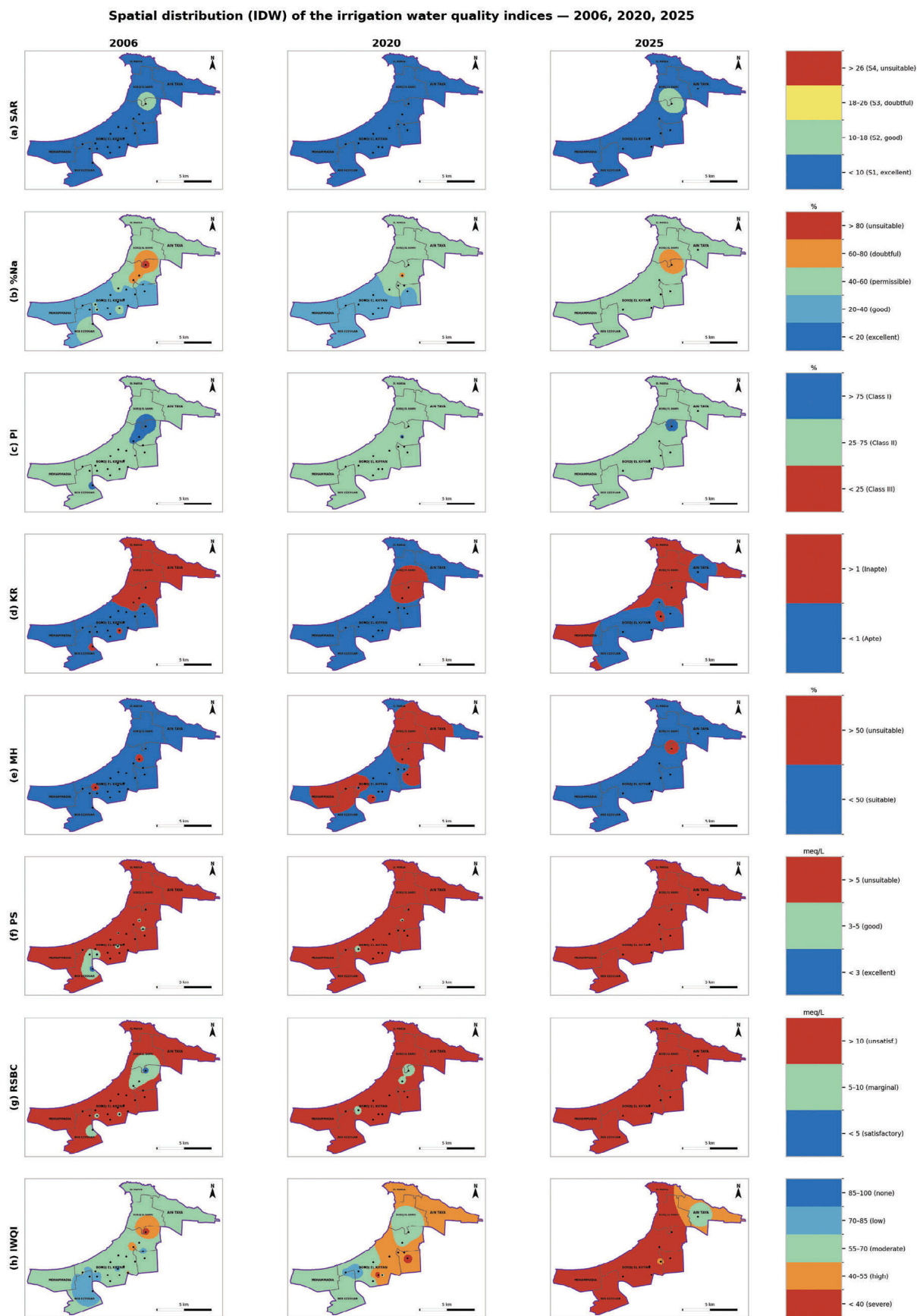


Fig. 3 - Spatial distribution of irrigation water quality indices in 2006, 2020, and 2025: (a) SAR, (b) %Na, (c) RSBC, (d) KR, (e) MH, (f) PS, (g) PI, and (h) IWQI.

Fig. 3 - Distribuzione spaziale degli indici di qualità per l'uso irriguo delle acque negli anni 2006, 2020 e 2025: (a) SAR, (b) %Na, (c) RSBC, (d) KR, (e) MH, (f) PS, (g) PI, e (h) IWQI.

water quality. The mean value thus shifted from moderate restriction in 2006 to moderate restriction near the lower limit in 2020, and subsequently to severe restriction in 2025. Spatially, sectors exhibiting moderate to high restrictions dominated in 2006, whereas more severe restrictions expanded in 2020. By 2025, severe restriction conditions became broadly dominant, particularly in Mohammedia, Bab Ezzouar, Bordj El Kiffan, and Bordj El Bahri, while less restrictive conditions persisted locally in El Marsa and Ain Taya. Categories corresponding to low or no restriction were no longer observed in 2025. This evolution highlights a clear spatiotemporal degradation of water suitability for irrigation, consistent with the increase in electrical conductivity, TDS, Na^+ , Cl^- , and RSBC. This pattern reflects an intensification of groundwater mineralisation (Meireles et al., 2010; Gaagai et al., 2023).

Hydrochemical Indices

The descriptive statistics of the main hydrogeochemical indices (CAI-1, CAI-2, r_1 and r_2) calculated for the 2006, 2020, and 2025 campaigns are presented in Table 5. The chloro-alkaline indices (CAI-1 and CAI-2) indicate changes in ion exchange within the aquifer. CAI-1 decreased between 2006 and 2020 before increasing in 2025, while CAI-2 showed the same general pattern with greater spatial variability (Table 5). According to Schoeller (1965), predominantly positive values are consistent with exchange of dissolved Na^+ and K^+ for Ca^{2+} and Mg^{2+} adsorbed on the aquifer matrix. Soltan ratios r_1 and r_2 also varied among

campaigns. Taken together, these indices are consistent with a combination of ion exchange, mineral dissolution and mixing; they do not, on their own, establish a temporal intensification of any single process.

Hydrochemical Facies and Controlling Mechanisms
Piper and Gibbs Diagrams

Figure 4 presents the Piper diagrams for the hydrochemical facies observed in 2006, 2020 and 2025. In 2006, groundwater predominantly exhibited a calcium-magnesium chloride-sulphate facies (Ca-Mg-Cl-SO_4), with only PZ58 and PZ59 displaying a sodium chloride facies (Na-Cl). In 2020, the Ca-Mg-Cl-SO_4 facies remained dominant, but PZ58 exhibited a Na-Cl facies and several points (PZ54, PZ32, PZ41) evolved towards it. In 2025, this evolution intensified. The general trend indicates an increasing influence of sodium and chloride, consistent with the CAI and Soltan indices. This salinisation likely reflects the combined effects of seawater intrusion and water-rock exchange typical of coastal aquifers (Piper, 1944; Appelo & Postma, 2005). These observations are consistent with the findings of Yahiaoui et al. (2023), Balasubramanian et al. (2022), Mahlknecht et al. (2017), and Firat Ersoy et al. (2021), who demonstrated that seawater intrusion in coastal aquifers is accompanied by sodium and chloride enrichment. The Gibbs diagrams (Gibbs, 1970) show that groundwater chemistry is primarily controlled by water-rock interaction and evaporation-crystallisation (Fig. 5). In 2006, most samples plotted within the transition zone between these two mechanisms, whereas in 2020 and especially in 2025, the points

Tab. 5 - Descriptive statistics of the hydrogeochemical indices (Min; Max; Mean $\pm$ SD).
Tab. 5 - Statistiche descrittive degli indici idrogeologici (minimo, massimo, media $\pm$ deviazione standard).

Index	2006			2020			2025		
	Min	Max	Mean $\pm$ SD	Min	Max	Mean $\pm$ SD	Min	Max	Mean $\pm$ SD
CAI-1	0.48	20.89	5.19 $\pm$ 5.19	-2.37	4.35	1.62 $\pm$ 2.27	3.40	56.63	27.74 $\pm$ 16.96
CAI-2	-315.37	14.70	-17.74 $\pm$ 77.99	2.32	14.18	8.72 $\pm$ 3.96	-65.51	57.63	14.20 $\pm$ 33.19
r_1	-329.32	6.28	-24.13 $\pm$ 81.99	0.24	22.76	8.02 $\pm$ 5.99	-105.31	35.29	-7.88 $\pm$ 39.11
r_2	-329.11	6.28	-24.09 $\pm$ 81.94	0.60	22.92	8.15 $\pm$ 5.95	-105.21	35.34	-7.77 $\pm$ 39.11

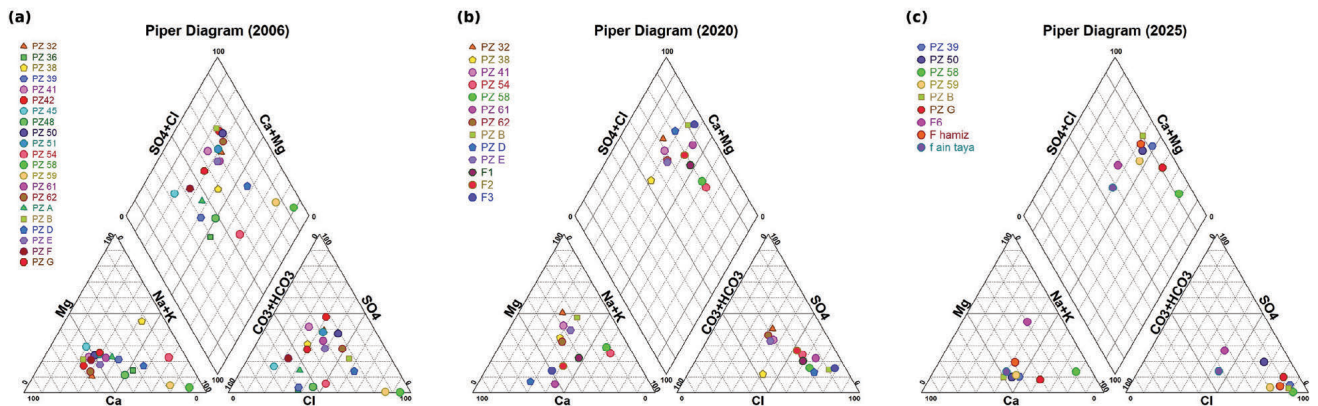


Fig. 4 - Piper diagrams illustrating the temporal evolution of the hydrochemical facies of groundwater in 2006, 2020 and 2025.
Fig. 4 - Diagrammi di Piper che illustrano l'evoluzione temporale delle facies idrochimiche delle acque sotterranee negli anni 2006, 2020 e 2025.

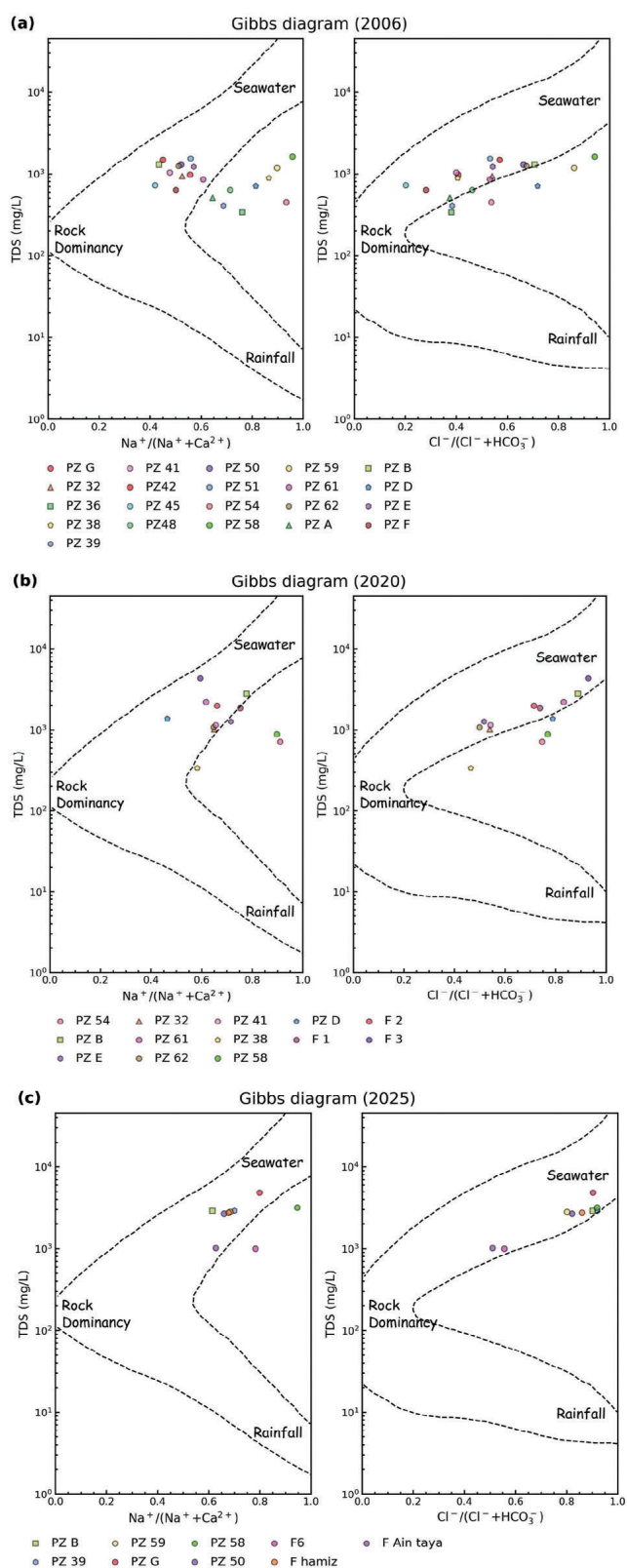


Fig. 5 - Gibbs diagrams for (a) 2006, (b) 2020 and (c) 2025. For each campaign, the left panel shows TDS versus $\text{Na}^+ / (\text{Na}^+ + \text{Ca}^{2+})$, and the right panel shows TDS versus $\text{Cl}^- / (\text{Cl}^- + \text{HCO}_3^-)$.

Fig. 5 - Diagrammi di Gibbs per l'anno (a) 2006, (b) 2020 e (c) 2025. Per ciascuna campagna, il pannello di sinistra mostra i solidi disciolti totali (TDS) vs $\text{Na}^+ / (\text{Na}^+ + \text{Ca}^{2+})$ e il pannello di destra mostra i solidi disciolti totali (TDS) vs $\text{Cl}^- / (\text{Cl}^- + \text{HCO}_3^-)$.

shifted progressively towards the evaporation–crystallisation domain, reflecting increasing mineralisation. Although the Gibbs diagram does not directly identify seawater intrusion, all hydrochemical indicators suggest an increasing influence of this phenomenon, in agreement with observations in other Mediterranean coastal aquifers (Gueddari et al., 2022). Thus, the Gibbs diagram confirms that the chemical composition of the waters results from the combined action of water–rock interactions, evaporation–crystallisation, and progressive salinisation linked to seawater intrusion.

Wilcox and USSL Diagrams

The USSL diagram (EC–SAR, Fig. 6) reveals a progressive deterioration of irrigation suitability over the 2006–2025 period, primarily driven by the increase in salinity. In 2006, most samples belonged to the C2-S1 and C3-S1 classes (moderate to high salinity, low sodicity), with the exception of PZ58 and PZ59 (C4-S2, C4-S3). In 2020, samples evolved primarily towards C3-S1 and C4-S1, confirming an increase in salinity. In 2025, most samples belonged to C4-S1 (very high salinity, low sodicity), with PZ58 evolving towards C4-S2 and PZG towards C4-S3. The Wilcox diagram (%Na–EC, Fig. 7) confirms this trend, with a progressive displacement of samples towards the “permissible” and “doubtful” domains. Overall, these diagrams show a progressive degradation of irrigation suitability, with salinity acting as the main limiting factor while sodicity, although generally low, increases locally in the most affected sectors.

Discussion

Hydrochemical evolution and controlling processes

Hydrochemical conditions between 2006 and 2025 show increasing mineralisation and salinity. Mean EC increased from 1,420 to 3,980 $\mu\text{S}/\text{cm}$, together with mean Na^+ and Cl^- concentrations. Na^+ and Cl^- were strongly correlated in all three campaigns ($r = 0.946, 0.959$ and 0.951 ; $p < 0.01$), supporting their joint contribution to the salinity gradient. Their strong correlations with EC in 2025 ($r = 0.921$ and 0.973 ; $p < 0.01$) are consistent with this interpretation. However, cross-campaign changes in correlation strength or significance cannot be attributed confidently to changing processes because the number of samples declined from 21 to 9. The hydrochemical interpretation therefore relies on the convergence of concentration patterns, facies, indices and spatial distributions, rather than on differences between individual correlation coefficients. The overall pattern is compatible with increasing marine influence (Yahiaoui et al., 2023; Morsli et al., 2017; Mahlknecht et al., 2017; Firat Ersoy et al., 2021; Balasubramanian et al., 2022; Gueddari et al., 2022).

However, mineralisation does not appear to result from marine influence alone. The increase in Ca^{2+} and, to a lesser extent, Mg^{2+} , together with the relative stability of HCO_3^- , suggests the persistence of water–rock interactions, particularly through carbonate dissolution (Appelo & Postma, 2005; Subba Rao et al., 2017). The behaviour of SO_4^{2-} also suggests a variable contribution from the dissolution of

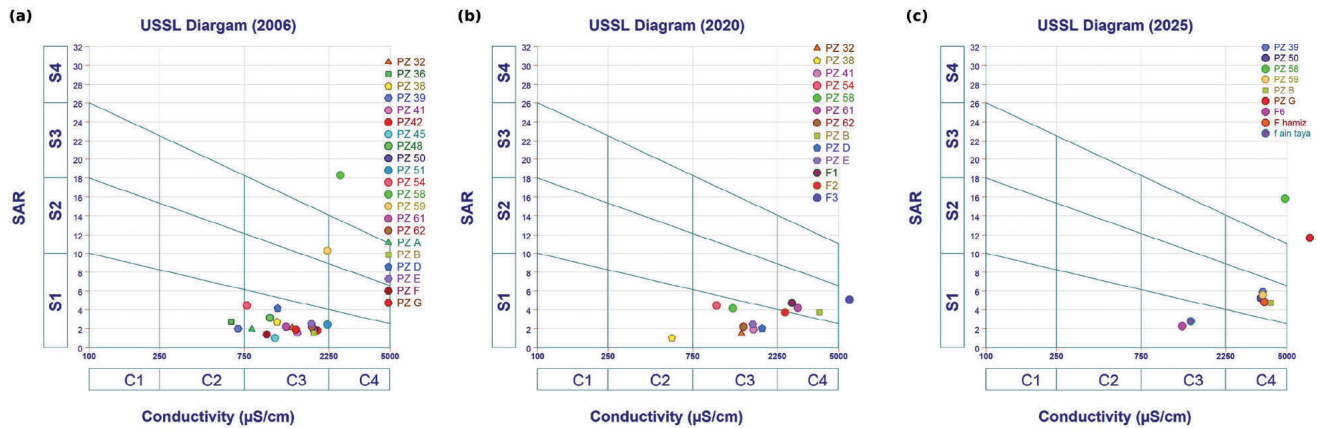


Fig. 6 - USSSL diagrams (EC–SAR) illustrating the temporal evolution of the suitability of groundwater for irrigation in 2006, 2020 and 2025.

Fig. 6 - Diagrammi USSSL (EC–SAR) che illustrano l'evoluzione temporale dell'idoneità delle acque sotterranee all'irrigazione negli anni 2006, 2020 e 2025.

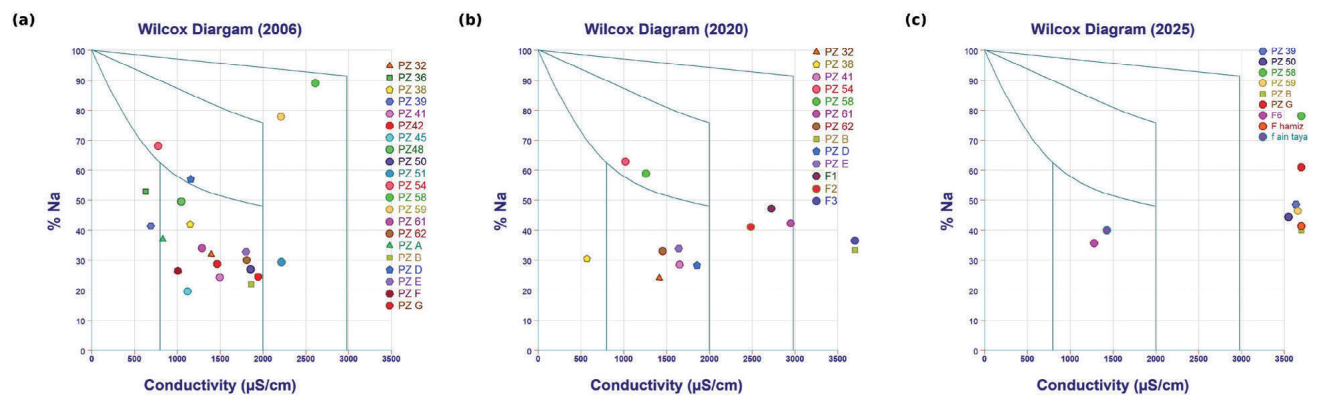


Fig. 7 - Wilcox diagrams (%Na–EC) illustrating the temporal evolution of groundwater suitability for irrigation in 2006, 2020 and 2025.

Fig. 7 - Diagrammi di Wilcox (%Na–EC) che illustrano l'evoluzione temporale dell'idoneità delle acque sotterranee all'irrigazione negli anni 2006, 2020 e 2025.

sulphate minerals and, potentially, anthropogenic inputs. The chloro-alkaline indices and Soltan ratios further demonstrate the contribution of ion exchange to the redistribution of Ca^{2+} , Mg^{2+} and Na^+ . Finally, the shift in facies in the Piper diagram, from Ca-Mg-Cl-SO_4 towards compositions increasingly dominated by Na-Cl , together with the movement in the Gibbs diagram towards the evaporation–crystallisation field, corroborates a progressive transformation of the hydrochemical system. The observed evolution therefore results from the combined effects of marine influence, water–rock interactions, ion exchange and concentration processes, with increasing salinisation over time.

Spatial distribution of deterioration and monitoring priorities

Groundwater deterioration shows pronounced spatial heterogeneity. The maps and diagrams show high EC, Cl^- , Na^+ , SAR, Kelley ratio, and potential salinity values associated with the lowest IWQI values, mainly along the coastal strip and in the eastern sectors, particularly at Bordj El Kiffan and Bordj El Bahri–Stamboul. Points PZ58, PZ59 and PZG are among the most affected and show signatures consistent with marine influence, characterised by Na-Cl facies in the Piper diagram and high-salinity classes in the USSSL diagram

(C4–S2 to C4–S3). In contrast, the inland and western sectors retain relatively better quality, more closely associated with water–rock interaction.

This spatial pattern identifies the coastal and eastern sectors as priorities for resource monitoring. Increasing the density of the monitoring network would improve documentation of salinisation trends and enable the early detection of further deterioration, particularly around the most affected points. It also highlights the importance of appropriately managing abstraction in vulnerable areas and strengthening the protection of inland and western zones, which retain relatively better quality. Spatial analysis therefore helps identify priority areas for monitoring and management, complementing the temporal assessment.

Implications for irrigation suitability

Hydrochemical deterioration is reflected in a progressive decline in irrigation suitability, with salinity representing the main constraint compared with sodicity. The decrease in IWQI from 64.82 in 2006 to 36.08 in 2025, together with the increase in the proportion of samples subject to severe restriction from 14.3% to 77.8%, demonstrates this trend. In 2025, severe restrictions predominated particularly at Mohammadia, Bab Ezzouar, Bordj El Kiffan and Bordj El

Bahri, whereas less restrictive conditions persisted locally at El Marsa and Aïn Taya. This trend is consistent with the increase in potential salinity, whose high values progressively spread across a large part of the study area.

Sodicity appears to be a secondary and more spatially localised constraint. Despite the increase in SAR and %Na in 2025, values corresponding to the most favourable categories remained predominant, while high values were concentrated particularly at Bordj El Bahri. The KR also indicated localised deterioration, mainly at El Marsa, Bordj El Bahri and Mohammadia. The increase in RSBC, whose values exceeded 10 meq/L for all samples in 2025, also indicates an unfavourable trend according to the criterion adopted in this study. MH remained generally favourable, despite locally high values at Bordj El Bahri and Bordj El Kiffan. Thus, salinity is the main limitation for irrigation, while the sodium-related risk tends to increase in certain sectors.

In this context, the IWQI is a relevant synthetic indicator for assessing the overall deterioration in irrigation suitability because it integrates several parameters representative of irrigation water quality (Meireles et al., 2010; Abbasnia et al., 2018; Gaagai et al., 2023). Persistent high salinity in areas subject to the most severe restrictions increases the risk of salt stress in sensitive crops and progressive soil salinisation. These findings therefore highlight the need for appropriate irrigation management and enhanced monitoring of the most affected sectors.

Management strategies to limit salinisation

In the region's semi-arid Mediterranean context, characterised by high evapotranspiration (970–1,250 mm/year), irregular precipitation and recurrent droughts, the results highlight the need for integrated and spatially targeted resource management. Reducing excessive abstraction, particularly in the most affected coastal and eastern sectors, is a priority for limiting declines in groundwater levels and slowing the advance of the saline wedge. Modernising irrigation systems and diversifying water resources, including the reuse of treated wastewater where its quality is appropriate, could also reduce dependence on groundwater.

At the same time, strengthening natural recharge and implementing managed aquifer recharge systems could help maintain groundwater levels and limit the advance of marine influence. In the eastern Mitidja Plain, areas associated with riverbeds, particularly those of the El Harrach and El Hamiz rivers, may offer opportunities for recharge through surface-water infiltration. Boufekane et al. (2020) identified an area with high recharge potential near the course of the Hamiz River, while Aziez et al. (2025) examined the feasibility of managed recharge solutions in the region. Such systems could include infiltration basins or check dams, subject to prior assessment of water quality, hydrogeological conditions and contamination risks. Thus, combining reduced abstraction, sound recharge management and strengthened piezometric and hydrochemical monitoring represents a relevant approach for slowing aquifer salinisation.

Novelty, broader significance and perspectives

The tools used in this study-IWQI, irrigation indices, Piper and Gibbs diagrams, multivariate statistics and GIS-are well established. The novelty lies in their integrated application to the coastal aquifer of the Bay of Algiers over almost two decades (2006–2025). The study jointly documents the temporal evolution and spatial redistribution of irrigation suitability, highlighting the intensification of salinisation and the most vulnerable sectors. It therefore complements the general groundwater-quality analysis of Ferdjouni et al. (2025) by focusing on hydrochemical processes, agronomic implications and the spatial dimension of deterioration. The results are consistent with regional observations that link groundwater deterioration to marine influence in the Mitidja Plain aquifer (Yahiaoui et al., 2023; Morsli et al., 2017; Aziez et al., 2025). Similar findings have been reported for other Mediterranean and arid coastal aquifers exposed to comparable pressures (Gueddari et al., 2022; Solangi et al., 2019). This integrated spatio-temporal approach strengthens the relevance of the results for monitoring and managing coastal aquifers exposed to increasing pressure.

The study has important limitations. The number of monitoring points decreased from 21 in 2006 to 13 in 2020 and 9 in 2025 because of urbanisation and well abandonment. This reduction decreases statistical power, makes correlation coefficients less stable and limits direct cross-campaign inference, especially for 2025. The temporal findings should therefore be viewed as campaign-level evidence rather than as paired-well trends. In addition, the assessment is based mainly on major-ion chemistry. Environmental isotopes and the HFE-D diagram (Giménez-Forcada, 2010; 2019) could better distinguish marine intrusion from other sources and processes in future work.

Conclusion

This study assessed the spatio-temporal evolution of groundwater quality in the coastal aquifer of the Bay of Algiers in 2006, 2020 and 2025. The integrated approach combined physicochemical parameters, irrigation suitability indices (IWQI, SAR, %Na, KR, PI, RSC, RSBC, MH and PS), hydrochemical tools (Piper, Gibbs, CAI-1, CAI-2, r1 and r2) and GIS-based spatial analysis. The results reveal a progressive deterioration in groundwater quality during the study period, mainly characterised by intensified mineralisation and salinisation.

The hydrochemical evolution shows a gradual transition from a system more strongly controlled by water–rock interactions in 2006 to one increasingly marked by salinisation in 2025. The increase in electrical conductivity and in the main ions associated with salinity, particularly Na⁺ and Cl⁻, together with the shift in facies towards Na–Cl compositions, demonstrates this transformation. Salinity is now the main constraint on irrigation, as illustrated by the decrease in IWQI and the increase in the proportion of samples subject to severe restriction, from 14.3% in 2006 to 77.8% in 2025. Sodicity indicators remain generally more

favourable, but their evolution and the presence of localised anomalies, particularly in the Bordj El Bahri, Bordj El Kiffan and El Marsa sectors, indicate the emergence of a spatially heterogeneous sodicity risk.

The deterioration therefore exhibits a contrasting spatial distribution, with the coastal and eastern sectors, particularly Bordj El Kiffan and Bordj El Bahri–Stamboul, appearing to be the most vulnerable, while the inland and western sectors retain relatively better quality. These findings highlight the need to prioritise enhanced monitoring in the most affected areas, control groundwater abstraction and promote sound aquifer-recharge management. Combining these measures with water-resource diversification and improved irrigation efficiency is essential to limit the progression of salinisation and sustainably preserve the groundwater resource and its agricultural use.

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Author contributions

Conceptualization, S.M. and M.Z.; methodology, S.M. and M.Z.; data collection and analysis, S.M.; interpretation of results, S.M., M.Z. and O.A.; writing-original draft preparation, S.M.; writing-review and editing, S.M., M.Z. and O.A.; visualization, S.M. All authors have read and agreed to the final version of the manuscript.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The authors declare no competing interest.

AI use declaration statement

During the preparation of this work, the authors used generative AI tools solely to improve the language and readability of the English text. No AI tools were used to generate, analyse, or interpret data, nor to produce scientific content or conclusions. After using these tools, the authors reviewed and corrected the content, including references, as necessary and assume full responsibility for the content of the publication.

Data availability

The data supporting this study are available from the National Agency for Water Resources (ANRH) of Blida and may be provided by the corresponding author upon reasonable request.

Supplementary Materials

The following supporting information can be downloaded at: www.acquesotterranee.net/acque/article/view/1031

Additional information

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